

Anomaly Detection & Time Series Assignment Answers

Question 1: What is Anomaly Detection? Explain its types (point, contextual, and collective anomalies) with examples.

Anomaly Detection is the process of identifying unusual patterns or observations in data that differ significantly from normal behavior.

Types of Anomalies:

1. Point Anomaly:

A single observation is abnormal compared to the rest of the data.

Example: A credit card transaction of ₹10 lakh from a customer who usually spends ₹5000.

2. Contextual Anomaly:

An observation is abnormal in a specific context.

Example: High electricity usage during midnight may be abnormal.

3. Collective Anomaly:

A group of observations together behaves abnormally.

Example: Multiple failed login attempts within a short period may indicate a cyberattack.

Question 2: Compare Isolation Forest, DBSCAN, and Local Outlier Factor in terms of their approach and suitable use cases.

1. Isolation Forest:

- Approach: Detects anomalies by isolating observations using random trees.
- Best Use Case: High-dimensional datasets and large-scale anomaly detection.

2. DBSCAN:

- Approach: Density-based clustering algorithm that identifies anomalies as low-density points.
- Best Use Case: Spatial data and clustering problems.

3. Local Outlier Factor (LOF):

- Approach: Measures local density deviation of a data point compared to neighbors.
- Best Use Case: Detecting local anomalies in datasets with varying densities.

Question 3: What are the key components of a Time Series? Explain each with one example.

The key components of a time series are:

1. Trend:

Long-term increase or decrease in data.

Example: Growth in online shopping sales over years.

2. Seasonality:

Repeating patterns over fixed intervals.

Example: Ice cream sales increasing every summer.

3. Cyclical Component:

Fluctuations occurring over irregular periods.

Example: Economic business cycles.

4. Residual/Noise:

Random variation that cannot be explained.

Example: Sudden unexpected stock market movements.

Question 4: Define Stationary in time series. How can you test and transform a non-stationary series into a stationary one?

A stationary time series has constant mean, variance, and autocorrelation over time.

Testing Stationarity:

- Augmented Dickey-Fuller (ADF) Test
- KPSS Test

Transforming into Stationary:

- Differencing
- Log transformation
- Seasonal differencing
- Detrending

Question 5: Differentiate between AR, MA, ARIMA, SARIMA, and SARIMAX models in terms of structure and application.

1. AR (AutoRegressive):

Uses past values to predict future values.

2. MA (Moving Average):

Uses past forecast errors to predict future values.

3. ARIMA:

Combines AR, MA, and differencing for non-stationary data.

4. SARIMA:

Extension of ARIMA that handles seasonal patterns.

5. SARIMAX:

SARIMA with external variables (exogenous variables) such as weather or promotions.

Question 6: Load a time series dataset (e.g., AirPassengers), plot the original series, and decompose it into trend, seasonality, and residual components.

```
from statsmodels.tsa.seasonal import seasonal_decompose
import pandas as pd
import matplotlib.pyplot as plt

data = pd.read_csv('AirPassengers.csv')

result = seasonal_decompose(data['#Passengers'], model='multiplicative', period=12)

result.plot()
plt.show()
```

Question 7: Apply Isolation Forest on a numerical dataset (e.g., NYC Taxi Fare) to detect anomalies. Visualize the anomalies on a 2D scatter plot.

```
from sklearn.ensemble import IsolationForest
import matplotlib.pyplot as plt
import pandas as pd

data = pd.read_csv('taxi_fare.csv')

model = IsolationForest(contamination=0.05)
data['anomaly'] = model.fit_predict(data[['fare_amount']])

plt.scatter(data.index, data['fare_amount'], c=data['anomaly'])
plt.show()
```

Question 8: Train a SARIMA model on the monthly airline passengers dataset. Forecast the next 12 months and visualize the results.

```
from statsmodels.tsa.statespace.sarimax import SARIMAX
import pandas as pd
import matplotlib.pyplot as plt

data = pd.read_csv('AirPassengers.csv')

model = SARIMAX(data['#Passengers'], order=(1,1,1), seasonal_order=(1,1,1,12))
result = model.fit()

forecast = result.forecast(12)
```

```
plt.plot(data['#Passengers'], label='Original')
plt.plot(range(len(data), len(data)+12), forecast, label='Forecast')
plt.legend()
plt.show()
```

Question 9: Apply Local Outlier Factor (LOF) on any numerical dataset to detect anomalies and visualize them using matplotlib.

```
from sklearn.neighbors import LocalOutlierFactor
import matplotlib.pyplot as plt
from sklearn.datasets import make_blobs
```

```
X, _ = make_blobs(n_samples=300, centers=1, random_state=42)
```

```
lof = LocalOutlierFactor(n_neighbors=20)
y_pred = lof.fit_predict(X)
```

```
plt.scatter(X[:,0], X[:,1], c=y_pred)
plt.show()
```

Question 10: You are working as a data scientist for a power grid monitoring company.

Real-Time Data Science Workflow:

1. Detecting Anomalies:

- Use Isolation Forest for large streaming data.
- Use LOF for local anomaly detection.
- Use DBSCAN for cluster-based anomaly detection.

2. Forecasting Model:

- Use SARIMAX because it can handle seasonality and external variables like weather.

3. Validation and Monitoring:

- Use RMSE, MAE, and MAPE metrics.
- Monitor drift and retrain models periodically.

4. Business Benefits:

- Prevents power failures.
- Improves energy distribution.
- Reduces operational cost.
- Helps in demand planning.

Sample Python Code:

```
from sklearn.ensemble import IsolationForest

model = IsolationForest()
model.fit(X_train)

print("Anomaly Detection Model Trained")
```